EHMbrAIn

AI-based versus traditional engine health monitoring:
a gas path analysis testbed on the CFM56-7B with synthetic ground truth

— anonymous in the public repository —

August 2, 2026

Where this talk goes

Two levels throughout: what the result is, then the mechanism that produces it.

The problem

The physics, and why diagnosis is hard

The testbed

What classical GPA delivers here

The pre-registered contest

Beyond the wall

What to take away

The problem

An engine degrades where nobody can look

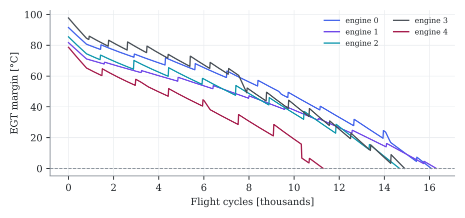
The whole discipline exists because the interesting state is unobservable.

A commercial turbofan stays on the wing for years. Meanwhile:

- dust and pollution foul the compressors,
- hot-section parts creep and oxidize,
- tip clearances open, seals wear.

Nobody sees any of it. The airline sees a handful of numbers per flight: shaft speeds, exhaust temperature, fuel flow.

The stakes: a shop visit costs millions; an unscheduled removal disrupts a whole fleet schedule.



The one aggregate an operator watches: exhaust-gas-temperature margin. Sawteeth are compressor washes; zero is retirement.

Two schools, and a broken comparison

The literature's structural weakness is the reason this project exists.

Traditional EHM — since the 1970s

Gas path analysis: invert a thermodynamic model to ask *which component* degraded and *by how much*. Around it: baselines, trend smoothing, Kalman filters, expert rules.

AI-based EHM — last decade

Striking results on remaining-life prediction, mostly on synthetic benchmarks (C-MAPSS and successors).

The gap

AI is almost never compared against a traditional pipeline *implemented with equal care*, on the same data, with the same metrics and the same tuning effort. A stream of comparisons against straw men tells us nothing about what AI actually contributes, when, or at what cost.

What this project builds

A level playing field, plus the one thing no real fleet can provide.

1. A **calibrated thermodynamic twin** of the CFM56-7B26, built only from certified public data.
2. **SynCFM56**: 100 virtual engines run to failure, in airline snapshot format, carrying complete ground truth — the true health of every component at every flight.
3. Both EHM families on *identical* data, metrics, splits and tuning budget, adjudicated under a **pre-registered protocol**.

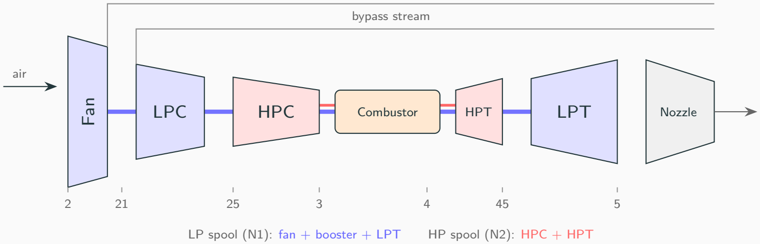
Why ground truth changes the question

On a real fleet you can ask "did the method predict well?". Only with synthetic truth can you ask "did it blame the right component?" — which is exactly where traditional GPA is weakest and AI is least transparent.

The physics, and why diagnosis is hard

The machine, in one picture

Two independent spools; the station numbers every later slide uses.



The airline receives N1, N2, EGT and fuel flow — the "cockpit set". Optional packages add pressures and temperatures at stations 25 and 3.

Gas path analysis: the model to invert

Ten unknowns, three usable measurements. That ratio is the whole story.

Health state — two numbers per turbomachine, in % off a new engine:

$$\mathbf{x} = [\Delta\eta_{\mathrm{fan}}, \Delta\Gamma_{\mathrm{fan}}, \ldots, \Delta\eta_{\mathrm{LPT}}, \Delta\Gamma_{\mathrm{LPT}}]^{\mathsf{T}} \in \mathbb{R}^{10}$$

η: isentropic efficiency (dirt and wear reduce it). Γ: flow capacity (fouling closes compressors; erosion opens turbines).

What is measurable — deviations from a healthy-engine baseline:

$$\Delta\mathbf{z} = \underbrace{\mathbf{H}(\mathbf{u})}_{\text{influence matrix}} \mathbf{x} + \mathbf{S}\mathbf{b} + \mathbf{v}, \qquad \mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$$

The inversion is ill-posed

With the cockpit set, $\mathrm{rank}(\mathbf{H}) = 3$ for 10 unknowns. Infinitely many health states explain the same measurements. Regularized weighted least squares picks one — and pays for it in bias.

Smearing: how a diagnosis flips to the wrong component

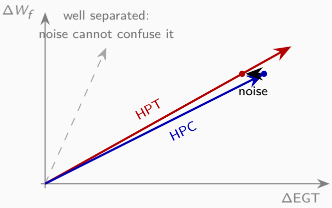
Two faults, two measurements, one noise-sized error. Nothing here is a bug.

Two health parameters, two channels: $\mathbf{H} = \left(\begin{smallmatrix} 1.0 & 0.9 \\ 0.5 & 0.5 \end{smallmatrix}\right)$.

Truth: HPC fault only, $\mathbf{x} = (-1, 0)$. Exact measurements invert perfectly.

Perturb one channel by $+0.1$ — one noise standard deviation — and the inversion returns $\hat{\mathbf{x}} = (0, -1)$:

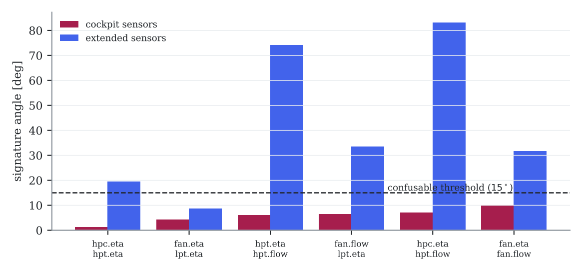
HPC healthy, HPT degraded. The information is not in the measurements when two signatures nearly coincide.



The two fault signatures sit 2.5° apart. One nudge moves the reading from one ray onto the other.

This engine's actual geometry

The wall is not hypothetical: it is measured, and it does not move.



With the **cockpit set**:

- minimum signature angle 1.3° (η_{HPC} vs η_{HPT}),
- seven pairs below 15°,
- rank 3 of 10.

With the **extended set** (P25/T25/PS3/T3): rank 7, minimum angle ×5.

Repeated over a six-point envelope grid, the structure barely moves (1.2–1.4°): the blindness is intrinsic to the sensor set, not to one flight condition.

The testbed

The twin: certified data in, predictions out

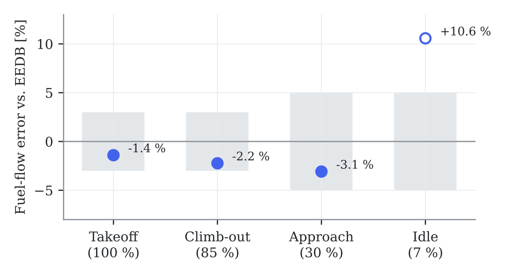
Calibrated against measurements it was never fitted to — that is the test.

Built in pyCycle by morphing a validated NASA example, at most three parameters per step, convergence required at each.

Calibration uses only **measured certified data**: type-certificate limits, and the ICAO databank's sea-level test-bed *fuel flows*.

Points are solved *thrust-matched*, so the model's fuel flow is a prediction confronted with the measurement — not a fitted quantity.

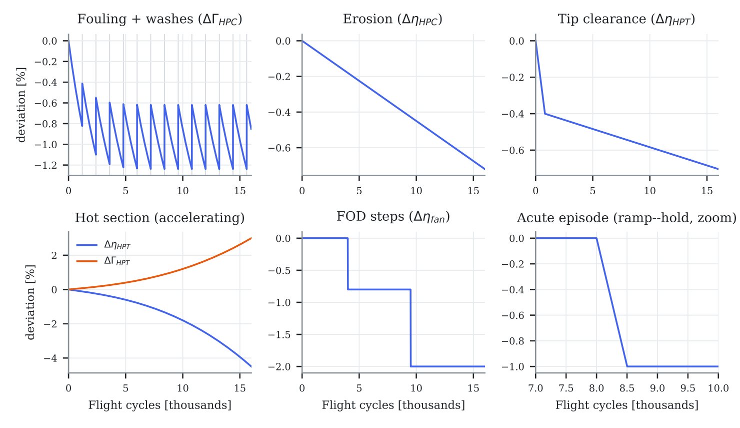
Result: takeoff $-1.4\,\%$, climb-out $-2.2\,\%$, approach $-3.1\,\%$. Cruise TSFC lands within $2\,\%$ of the published value *without being a target*.



Gray bands are the contract tolerances, fixed *before* the run. The hollow marker (idle) is reported but not gated — deep part power is generic-map territory.

The fleet: six mechanisms, one hundred lives

Each mechanism is a closed-form function of flight cycle, stored separately.



Drawn by the generator's own code. Each contribution is stored apart, so the fleet records not just *how* degraded an engine is but *by what*.

Making the benchmark hard on purpose

A synthetic dataset's commonest disease is being accidentally easy.

The difficulty gate

A deliberately weak classifier (logistic regression on raw channels) must *fail* to solve the isolation task. Threshold set in advance: 60 %.

First pass: failed at 81.6 %.

Root cause: chronic mechanisms co-evolve with age, so the label was an age proxy, not a fault label. The task was never isolation.

Fix was *structural*, not cosmetic: discrete single-parameter acute fault episodes, drawn from the confusable pairs.

Second pass: 54.0 % — pass.

And on fleet v1.1 the gate bit again (62.2 %): fault magnitudes were scaled by 0.75 until it passed at 58.1 %.

An audit also caught a generator bug: episode onsets drawn against the simulation horizon instead of engine life.

Both failures are in the report in full. A gate that cannot fail certifies nothing.

Four safeguards, fixed before any contestant ran

The comparison is itself an instrument, and it was designed first.

Pre-registration. Hypotheses, thresholds, splits (with file hashes), tests and stopping rule frozen under a git tag before the confirmatory run.

Symmetric tuning. 50 Optuna trials *each*. The comparison is between two tuned practitioners, not a tuned one and a default one.

Split by engine, 70/10/20, never by flight — flights of one engine are near-duplicates. Leakage checked in CI.

A strong baseline. The traditional pipeline is built to industrial shape, every expert rule justified; smearing is *measured* against truth, not assumed.

Evidence hierarchy

Numbers exist in two castes and are never mixed: *exploratory* (produced while building) and *confirmatory* (produced once, after the freeze, on data touched exactly once). Only the second supports a verdict.

What classical GPA delivers here

The experiment: honest forward, classical inverse

Inverting the matrix that generated the data would be circular.

Forward (plays the engine)

The nonlinear neural twin: full physics curvature, 13–19$\times$ closer to pyCycle than a linearization.

Inverse (plays the monitoring system)

The linear regularized WLS on the influence matrix, at registered defaults — and, like every fielded GPA, unaware the physics is curved.

One run, end to end

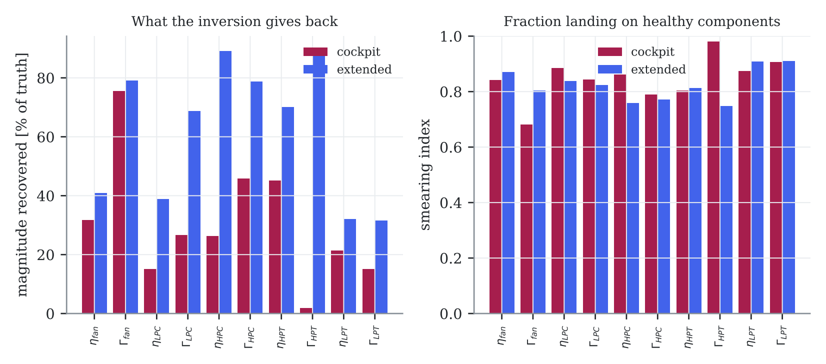
HPC loses 1 % efficiency. The cockpit sees N2 $-0.12\,\%$, W_f $+0.57\,\%$, EGT $+0.52\,\%$.

The inversion returns $-0.26\,\%$ on HPC efficiency — a quarter of the truth — and scatters the rest.

Over 500 noise draws it names the HPC correctly 22 % of the time. Favourite wrong answer: the HPT.

What the inversion gives back

Recovery and smearing, per parameter, per sensor set.

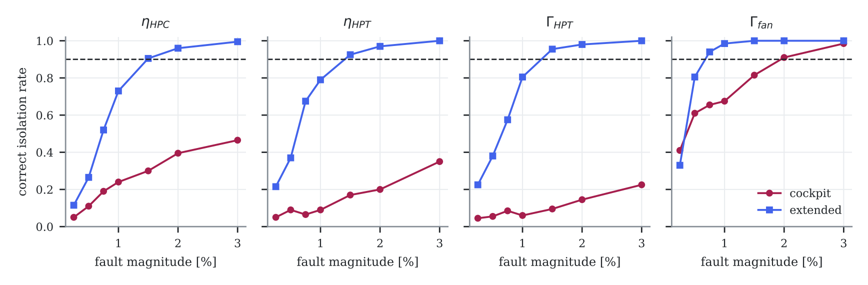


Cockpit: 2–76 % recovered, median smearing 0.85 — six sevenths of the diagnosed magnitude lands on healthy components.

Extended: 32–90 %, smearing barely moves (0.82). More sensors buy *magnitude*, not tidiness.

How small a fault can be named

The practical resolution limit of each sensor package.

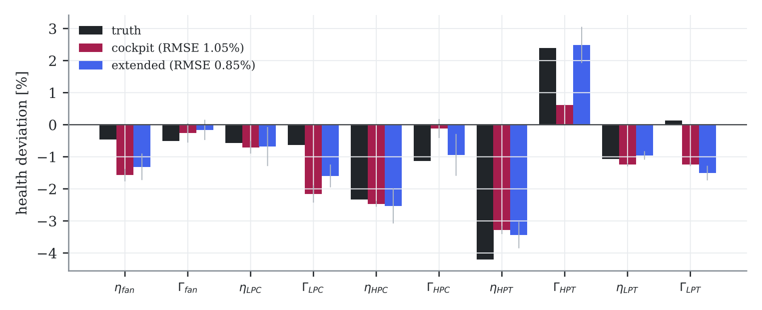


At the 90 %-correct criterion, the cockpit set resolves one parameter of ten; the extended set resolves seven, several from 0.75–1.5 %.

The fleet's own faults are 0.3–1.1 %. Most faults in this benchmark sit below the cockpit set's resolution limit — which is why the isolation contest later ends where it does.

A real engine, eight faults at once

Single-fault studies are the exercise; this is the situation.



True state: HPT efficiency $-4.2\,\%$ with flow capacity opened $+2.4\,\%$. The cockpit diagnosis gets the headline (HPT -3.3) but returns $+0.6$ for the flow opening — the eroded-nozzle fingerprint is absent. Both sets also invent deterioration: fan efficiency three times its true value, LPT flow wrong in sign. A decision on those two numbers opens a healthy module.

The pre-registered contest

Five hypotheses, one confirmatory pass

Three confirmed, two refuted. The split is the message.

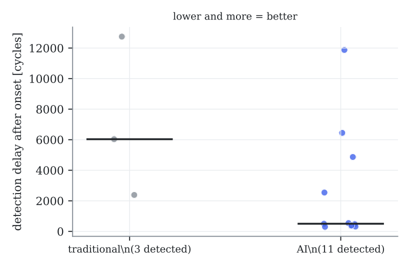
Hypothesis	Verdict	Confirmatory evidence (AI vs traditional)	p (Holm)
H1 (detection)	CONFIRMED	recall 0.48 vs 0.13; delay 499 vs 6033 cy; FA 1=1	0.008
H2 (confusable isolation)	REFUTED	0.31 vs 0.31 (n=13; discordant 4/4)	0.64
H3 (RUL prognosis)	CONFIRMED	RMSE 1694/858 vs 3816/1981 cy (50/90 % life); δ=0.89	8.6e-06
H4 (physics hybrid)	REFUTED	hybrid worse at 10/25/100 % (2852 vs 1772 cy at 10 %)	—
H5 (uncertainty)	CONFIRMED	coverage 0.883 vs 0.983 (nominal 0.90); half-width 1957 vs 11509 cy	—

Both families tuned under the frozen 50-trial budget; the 20 test engines evaluated exactly once; Wilcoxon paired by engine, one-sided exact McNemar for binary outcomes, Holm correction, BCa intervals.

The two refutations did more work than the three confirmations: H2 turned an algorithm question into a purchase order, H4 removed a popular technique from the shortlist.

H1 — the story is the tuning, not the model

An honest budget moved the answer; that is itself the finding.



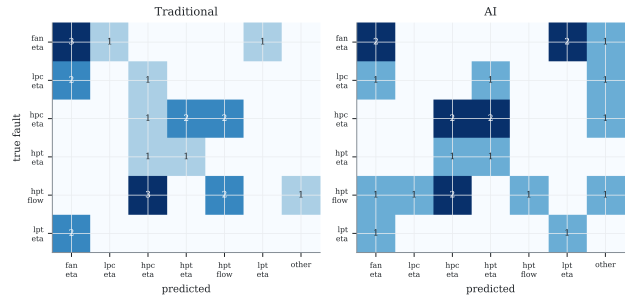
Untuned, the AI detector scored recall 0.06 and the report called it "a design gap, not a threshold gap".

The frozen search space contained the suspected lever — feature window length. Fifty trials stretched the windows: recall 0.48 at a 499-cycle median delay, twelve times earlier than the tuned classical detector, at equal false alarms.

Two transferable lessons: window length is the first knob any change detector lives or dies by; and threshold selection on small clean validation samples is fragile for *any* method.

H2 refuted — the wall binds everyone

Predicted by geometry before any model ran, and confirmed by both families.



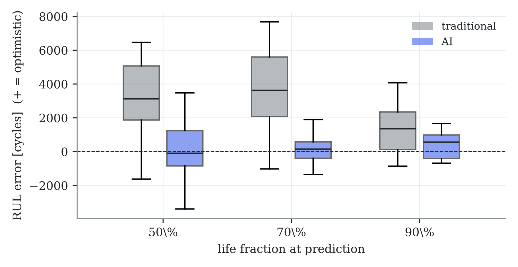
On the thirteen confusable test episodes the families tie exactly: 31 % each, with a perfectly symmetric disagreement pattern (four episodes only the AI gets, four only the rule gets).

The conclusion is hardware

If confusable isolation matters operationally, add gas-path instrumentation. No amount of modelling recovers information the sensors never captured.

H3 and H5 — prognosis is where learning pays

Not just a lower error: a differently shaped error.



RMSE 1694/1042/858 cycles at 50/70/90 % of life, against 3816/4544/1981. Cliff's $\delta = 0.89$, Holm $p = 8.6 \times 10^{-6}$.

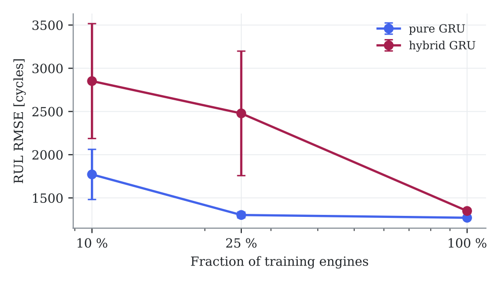
Two scales, and only one is a sales pitch: $\sim$2.3$\times$ over the extrapolation an operator fields *today*; over the best *advanced* classical prognostic, 1.34$\times$ at 90 % of life and a tie at 70 % ($p = 0.41$). The learning advantage is late-life.

Traditional errors are wide and skewed optimistic — the dangerous direction; the AI's are tight and near-unbiased.

H5: $\pm$1957 vs $\pm$11 509 cy — but the classical

H4 refuted — physics features are not free wins

The popular default of concatenating estimator outputs is a documented negative.



Feeding the Kalman-tracked health parameters to the network alongside the raw deviations made it worse at every data fraction — and *most* worse exactly where the hypothesis predicted it would help most.

Mechanism, in hindsight: those estimates are heavily smeared. They add ten noisy, mutually correlated channels whose information is already present in the three deviations they were computed from.

Refuted a second, independent way later, with twin-residual features on a nonlinear fleet.

Does the ranking survive outside our own simulator?

A benchmark this project had no hand in generating.

Re-ran the prognosis question on NASA's **C-MAPSS FD001**, the field's reference turbofan run-to-failure benchmark, with the same method shapes and three seeds.

Result

RMSE 14.9 cycles (AI) versus 42.1 (traditional) — same direction, comparable magnitude of advantage, and an absolute figure competitive with published results.

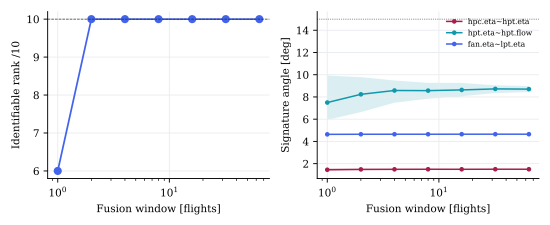
Disclosed substitution: the plan named N-CMAPSS, whose multi-gigabyte distribution breaks the project's desktop-reproducibility norm. FD001 answers the same ranking question at 5 MB, and only prognosis transfers (FD001 has no event labels).

The prognosis conclusion is not an artifact of SynCFM56.

Beyond the wall

Can observability be bought without hardware?

The influence matrix changes with flight condition — so the schedule is a design variable.



Free ambient and power scatter restores algebraic rank immediately, but signature angles saturate: free variation is real but thin.

Two cheaper exits than new probes:

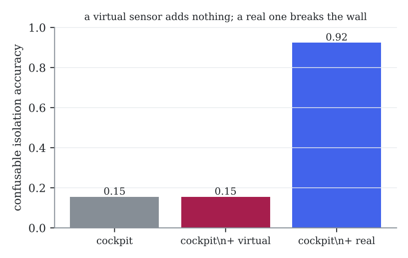
1. Redesign the report schedule. One added periodic low-power cruise report buys $+79\,\%$ separability on the HPT ambiguity. The cost is a procedure bulletin, not avionics.

2. Learn the fusion, keep the physics. A network fed per-flight *physics inversions* isolates at 0.65 against classical stacking's 0.17; fed raw sequences instead, it fails outright at 0.12.

What no schedule fixes: $\eta_{\mathrm{HPC}} \sim \eta_{\mathrm{HPT}}$ stays below 1.8° everywhere. That pair is fundamental.

Only a real sensor breaks the wall

The data-processing inequality, made operational.



Three sensor conditions on the confusable episodes:

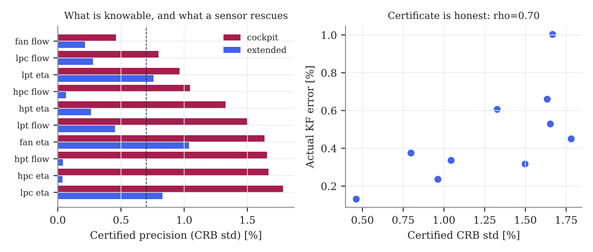
- cockpit only: 0.15
- cockpit + *model-predicted* station channels: 0.15 ($p = 1.000$)
- cockpit + *real* station channels: 0.92 ($p = 0.001$)

A quantity computed from the cockpit data cannot carry information the cockpit data did not hold. No software vendor can substitute a virtual sensor for a missing measurement.

Cheap in practice: PS3, T25 and T3 are already measured by the FADEC for control. The barrier is recording and downlinking them, not new hardware.

A certificate of what a diagnosis can know

And — uniquely — a proof that the certificate is honest.



Accumulate Fisher information over an engine's *own* flown conditions:

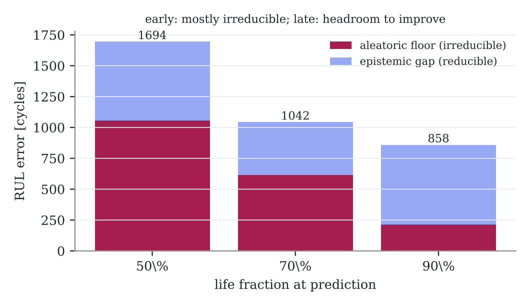
$$\mathbf{F} = \mathbf{P}_0^{-1} + \sum_t \mathbf{H}(u_t)^\top \mathbf{R}^{-1} \mathbf{H}(u_t)$$

The Cramér–Rao bound $\mathbf{F}^{-1}$ states the best per-direction precision *any* unbiased estimator can reach from this engine's data.

The proof (right panel): certified precision ranks the filter's *actual* error against ground truth, $\rho = 0.70$, $p = 0.025$ — but the control, added later: ≈ 0.24 is the shared matrix, ≈ 0.4

Is a large RUL error the method's fault, or the future's?

Different problems, opposite remedies. Confusing them wastes budget.



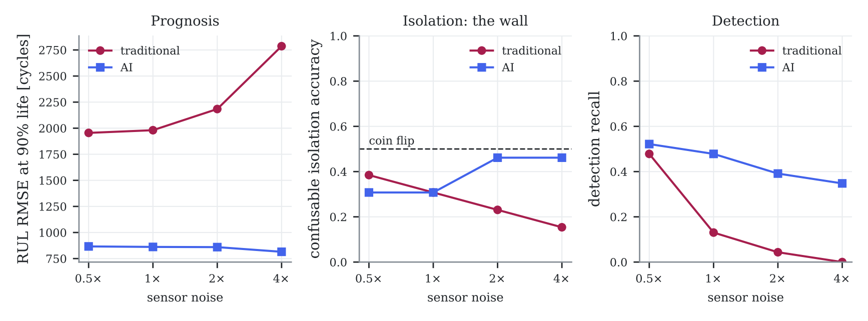
Condition on each engine's *true* current health, find its nearest neighbours in health space, and measure the spread of their *true* remaining lives. Two engines in the same state that fail hundreds of cycles apart differ for reasons no present measurement contains.

At mid-life: 87 % of the RUL spread is irreducible. Remaining life is, to first order, not written in the present.

Late in life: a $4\times$ reducible gap opens. That is where a better prognostic is worth building — and only there.

The determinability map: when does each approach win?

Sensor quality changes who wins; sensor placement changes what is knowable.



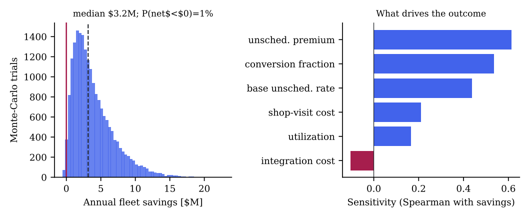
Prognosis is noise-proof. The AI's error is flat across a factor of eight in noise; the advantage widens from 2.3× to 3.4×.

The wall is geometric. Halving every σ leaves confusable isolation at 0.31–0.38. Buy new stations, not better probes at the old ones.

Thresholds fall off a cliff. Classical detection goes from 0.48 recall to zero; the AI holds 0.35 at quadruple noise.

What it is worth — and why it might be nothing

One mechanism monetized, because it is the only one the results support.



100 CFM56-7B engines. The only monetized mechanism: less-biased prognosis converting *unscheduled* removals into *scheduled* ones.

Median $3 M/year, p10 $0.9 M, p90 $8 M, net-negative in 1 % of draws. Read the spread, not the median.

Why it could be zero: the isolation wall is unbroken; false alarms erode trust; the sim-to-real gap preserved the ranking but not the magnitude; mature operators already catch most failures — and the absolute rate is anchored to the *industry* band, not to our fleet's optimistic raw numbers.

What to take away

The split verdict

Neither family wins outright, and the pattern is the useful part.

Learning wins where the task is *statistical aggregation over time*:

- prognosis: 858 vs 1981 cycles ($\sim$2.3$\times$ over fielded practice), robust across architectures and sensor quality — but only 1.34$\times$ over the best classical method, and only late in life
- detection timing: 12$\times$ earlier at equal false alarms
- calibrated uncertainty: 1.34$\times$ tighter at equal coverage — the 6$\times$ headline was mostly conformal, which is free to both sides

Nobody wins where the task is bounded by *what the sensors observed*:

- confusable isolation: $31\,\% = 31\,\%$
- the fix is +77 percentage points from *real* station probes
- virtual sensors add exactly nothing

And physics did not lose — it diagnosed the limits

The influence-matrix geometry predicted, *before any learning ran*, where every method would

What an engineering organization should do on Monday

In order of how much money each moves.

1. **Deploy learning for prognosis first.** The clear business case: maintenance slotting and lease planning, with calibrated intervals.
2. **Detection: modern change detection with tuned windows.** The win came from window search — cheap and auditable — not from depth.
3. **Isolation: buy sensors, not models.** And specifically probes at *new stations*; quieter probes at the old ones do not help.
4. **Redesign the report schedule before buying anything.** One procedural report condition recovers most of what deliberate spread offers.
5. **Protect the sensor estate.** A drifting thermocouple corrupts every consumer downstream, learned or classical.
6. **Distrust untuned comparisons in either direction** — both families moved dramatically under the same budget; scoreboards inverted.
7. **Demand pre-registration of vendor claims.** A split verdict is what honest evaluation looks like; a clean sweep should raise suspicion.

What transfers off this simulator — and what does not

Stating this precisely is the price of using synthetic data.

Transfers: rankings, mechanisms and geometry. Which method wins where and why; the observability structure any CFM56-class cockpit set shares; the sensor-acquisition logic.

Every operational claim in this work is of that kind.

Does not transfer: absolute performance numbers. An RMSE of 858 cycles is a property of this simulated world and its noise settings.

Honest limits: synthetic and linearized world (audited to the noise floor); two snapshot conditions with scatter; EGT is a station-45 proxy; one compact architecture per task.

Reproducibility

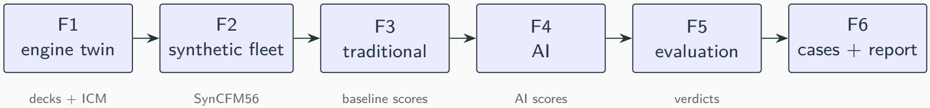
The entire evidence base regenerates in about eleven minutes on one desktop machine, GPU stages included. Every figure and number in the report and in these slides comes from a script — none is copied by hand.

The split verdict is the result.

A method comparison that produces a clean sweep

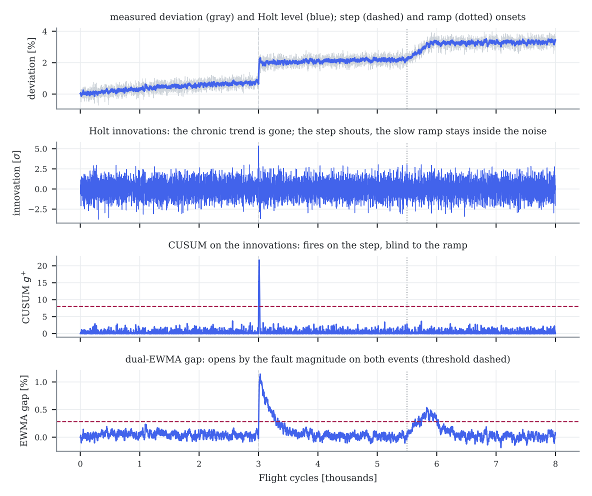
is usually measuring effort, not method.

Appendix — the pipeline



Each phase ends in a pass/fail gate with a criterion stated in advance. Extensions beyond F6 follow the same discipline: operating-point tomography, a ten-line limitation-overcoming programme, two ground-truth-validated certificates, and the economic study.

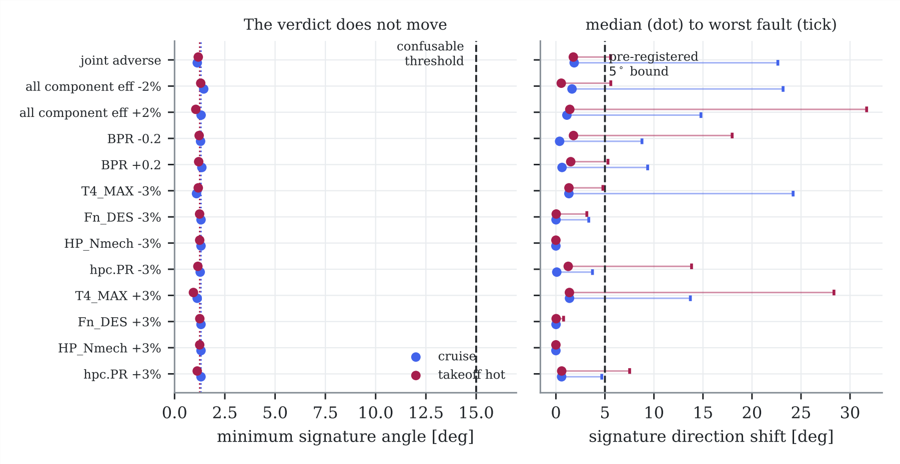
Appendix — the traditional detection stack



Step detectors belong on *innovations*: on raw deviations the chronic trend integrates without bound and alarms every engine.

And a trend-shift detector cannot see acute ramps — only the dual-EWMA gap does.

Appendix — is the geometry robust to calibration?



Every calibration knob perturbed inside its own contract tolerance. The verdict that matters is invariant: rank 3, minimum angle 0.94–1.45°, same most-confusable pair every time. What is *not* invariant is the exact membership of the confusable list at its 15° boundary — the weakest columns are near-null vectors, and a near-null vector has no stable direction.

Appendix — the compute budget

Stage	Script	Device	Wall time [s]
design point	`scripts/run_design_point.py`	CPU	2
sls anchors	`scripts/run_anchors.py`	CPU	13
baseline decks	`scripts/make_decks.py`	CPU	214
corrected baseline	`scripts/make_corrected_baseline.py`	CPU	1
icm grid	`scripts/make_icm.py`	CPU	71
fleet generation	`scripts/make_fleet.py`	CPU	40
audit dataset	`scripts/audit_dataset.py`	CPU	12
audit nonlinearity	`scripts/audit_nonlinearity.py`	CPU	20
traditional pipeline	`scripts/run_trad.py`	CPU	106
ai suite	`scripts/run_ai.py`	CPU+GPU	27
hybrid ablation	`scripts/run_hybrid.py`	CPU+GPU	104
pcs	`scripts/run_pcs.py`	CPU+GPU	46
surrogate data cruise	`scripts/f8_surrogate_data.py`	CPU	466
surrogate train	`scripts/f8_surrogate.py`	CPU+GPU	40
fleet v2	`scripts/make_fleet.py`	CPU	72
h4 v2 hybrid	`scripts/f8_l6_hybrid.py`	CPU+GPU	82
l4 recoverable	`scripts/f8_l4_recoverable.py`	CPU	4
l5 architectures	`scripts/f8_l5_arch.py`	CPU+GPU	52
l7 drift	`scripts/f8_l7_drift.py`	CPU	8
l9 pcs	`scripts/f8_l9_pcs.py`	CPU+GPU	96
identifiability certificate	`scripts/f10_certificate.py`	CPU	8
Full pipeline			**1484**

Reference machine: an Apple M5 desktop chip, 10 cores, 16 GB, tensor training on its GPU through the torch MPS backend — consumer hardware, deliberately. If it only ran on a cluster, it would not be adoptable.